

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Analyzed and designed geotechnical solutions for foundations, retaining structures, and slope stability across major infrastructure projects, including data centres and transportation networks.
- Prepared comprehensive **geotechnical design reports**, technical specifications, and desk studies for diverse developments including commercial and industrial sites.
- Managed, supervised, and interpreted results from multiple **geotechnical ground investigations**, providing critical technical support during construction phases.
- Integrated **digital tools and automation** into project workflows, enhancing efficiency in geotechnical analysis and design delivery by an estimated **15%**.
- Developed project management and commercial awareness through active participation in project planning and client interactions, improving project delivery timelines.
- Collaborated with cross-functional engineering teams to ensure the successful implementation of **ground engineering solutions**, aligning with sustainability objectives.

SKILLS

Geotechnical Engineering – Geotechnical design, Foundation analysis, Retaining structures, Slope stability, Earthworks, Soil mechanics, Ground investigations

Software & Tools – AutoCAD, Civil 3D, GIS, Numerical analysis software, Python, R

Reporting & Documentation – Design reports, Technical specifications, Desk studies, Site walkovers

Project Management – Planning, Organisation, Time management, Commercial awareness, Client liaison

Data Analysis & Automation – Data interpretation, Digital tools, Automation, Statistical analysis

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Infrastructure Stability Analysis Platform

- Developed a Python-based analytical tool for assessing slope stability and earthworks parameters, integrating with GIS data for comprehensive site analysis.
- Designed and implemented algorithms for calculating bearing capacity and settlement predictions for various foundation types, improving design accuracy by **10%**.
- Generated automated reports summarizing geotechnical findings and recommendations, streamlining communication with project stakeholders.

Digital Ground Investigation Management System

- Engineered a system to manage and interpret data from geotechnical ground investigations, incorporating numerical analysis for complex soil behaviours.
- Supervised the planning and execution of site investigations, ensuring adherence to technical specifications and safety standards.
- Contributed to the development of technical specifications for drilling and testing methodologies, enhancing data quality for subsequent analysis.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)